Home / Grade 10 / Geometry / Quarter 4

Control work 4, and work on the mistakes

Coordinate geometry in one paper, then the plane methods drawn as a single map.

LESSON 61–62 | 2 × 40 MIN | GEOMETRY 10, NAZORAT ISHI 4 | P1 3 REVIEW

LESSON CLOCK

3 | 42' | 12' | 23' | 10'

Instructions	3 min
The paper	42 min
Answers	12 min
Diagnosis and rewrite	23 min
The map	10 min

LEARNING OBJECTIVES

- Apply the line, circle and vector methods of Lessons 53–60 under time.
- Choose between the algebraic and the geometric route to a tangent.
- Classify each lost mark and rewrite the solution in full.
- Draw the coordinate block as one map.

TEXTBOOK

National	pp. 253–256
Cambridge	Pure Mathematics 1, pp. 73–76
Workbook	Control paper G4

01 Explanation

The paper — 35 marks, 42 minutes

Q	TASK	MARKS	FROM
1	$A(-1, 2)$, $B(7, 8)$: find $\lvert AB \rvert$, the midpoint and the gradient	5	L53–54
2	The same points: find the perpendicular bisector of AB	5	L53–54
3	Find the centre and radius of $x^2 + y^2 - 8x + 2y - 8 = 0$	5	L55–56
4	Find where $y = x - 1$ meets that circle, and the length of the chord	7	L57–58
5	Find the values of c for which $y = 2x + c$ is a tangent to $x^2 + y^2 = 45$	6	L57–58
6	Show that $P(1, 2)$, $Q(4, 8)$ and $R(7, 14)$ are collinear, using vectors	7	L59–60

WHERE THE MARKS ACTUALLY GO

Q2 carries one mark for the negative reciprocal and one for using the midpoint; Q3 one for the sign of the centre; Q4 one for giving both coordinates of each point; Q6 one for the concluding sentence. Five of the thirty-five marks are for a habit, not a calculation.

Naming the slip

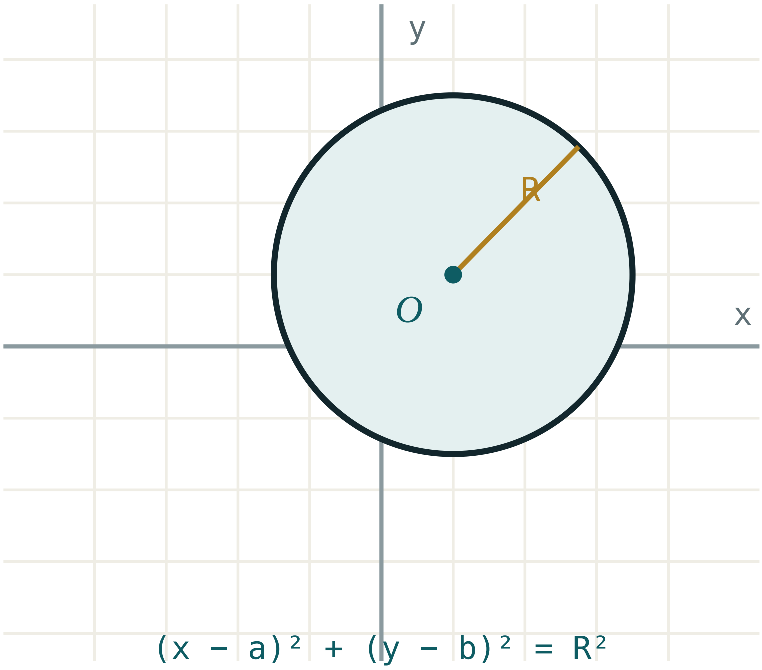
SLIP	WHAT IT LOOKS LIKE	THE FIX
gradient inverted	$\dfrac{x_2 - x_1}{y_2 - y_1}$	rise over run, in that order
centre sign	$(x - 3)^2 \Rightarrow$ *centre* -3	reverse the bracket sign
half a point	$x = 3,\ x = -4$	substitute back for y
bisector not through M	right gradient, wrong line	two conditions, both needed
tangent by guessing	no D and no distance	either route, but write one
vector proof unfinished	$QR = 2 \cdot PQ$	add “and Q is common”
AB reversed	$OA - OB$	“to minus from”

Name the slip in the margin, then rewrite the whole question. A rewritten line is forgotten by Friday; a rewritten solution is not.

The coordinate block as one map

Six boxes, links written as sentences:

- **two points** → **length, midpoint, gradient** — “three formulas, one subtraction each”
- **gradient** → **parallel and perpendicular** — “equal, or negative reciprocal”
- **midpoint + perpendicular** → **perpendicular bisector** — “equidistant from both”
- **centre and radius** → **circle equation** — “the distance formula, squared”
- **substitute** → **quadratic** → **discriminant** — “two, one or no points”
- **vector** → **parallel + common point** — “collinear”



The centre-and-radius picture that half the paper rests on.

TWO ROUTES TO A TANGENT, AND WHEN TO USE EACH

The **discriminant** route always works but is longer. The **distance** route, $\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}} = r$, is one line — use it whenever the circle's centre is easy to write down.

Looking forward

Lessons 63–68 leave the plane and return to space: parallelism and perpendicularity, sections of polyhedra and orthogonal projection, then the annual paper. Nothing new is introduced — it is Quarters I to III, revisited with a year's practice behind them.

ONE HABIT TO CARRY FORWARD

Coordinate geometry rewarded drawing the points before calculating. Solid geometry rewards drawing the solid before naming the angle. In both, the sketch is where the marks are decided.

02 Worked examples

EXAMPLE 1 **Model answer, Q2: the perpendicular bisector of $A(-1, 2)$ and $B(7, 8)$.**

1. $M = (3, 5)$

2. $m\_\{AB\} = \frac{6}{8} = \frac{3}{4}$

3. $m = -\frac{4}{3}$

   Negative reciprocal.

4. $y - 5 = -\frac{4}{3}(x - 3) \Rightarrow 4x + 3y = 27$

ANSWER

$4x + 3y = 27$

EXAMPLE 2 **Model answer, Q4: where $y = x - 1$ meets $x^2 + y^2 - 8x + 2y - 8 = 0$, and the chord length.**

1. Centre $(4, -1)$, $r = 5$.

   From Q3.

2. $x^2 + (x - 1)^2 - 8x + 2(x - 1) - 8 = 0 \Rightarrow 2x^2 - 10x - 9 = 0$

3. $x = \dfrac{10 \pm \sqrt{172}}{4}$, so $x \approx 5.78$ and $x \approx -0.78$.

4. Distance from the centre to the line $\dfrac{|4 - (-1) - 1|}{\sqrt{2}} = \dfrac{4}{\sqrt{2}}$; chord $2\sqrt{25 - 8} = 2\sqrt{17} \approx 8.25$.

   The faster route.

ANSWER

Chord $2\sqrt{17} \approx 8.25$

EXAMPLE 3 **Model answer, Q5: for which c is $y = 2x + c$ a tangent to $x^2 + y^2 = 45$?**

1. $r = \sqrt{45} = 3\sqrt{5}$

2. Distance from $(0, 0)$ to $2x - y + c = 0$ is $\dfrac{|c|}{\sqrt{5}}$.

3. $\dfrac{|c|}{\sqrt{5}} = 3\sqrt{5} \Rightarrow |c| = 15$

4. $c = \pm 15$

ANSWER

$c = 15$ or $c = -15$

03 Interactive model

Work Q5 both ways on the board — discriminant and distance — and time each; the class will not use the long one again.

Ten questions on the coordinate block

Gradient through $(1, 2)$ and $(5, 10)$:

- 2
- $\frac{1}{2}$
- 8
- 4

Perpendicular to gradient $\frac{3}{4}$:

- $\frac{4}{3}$
- $-\frac{4}{3}$
- $-\frac{3}{4}$
- $\frac{3}{4}$

Centre of $(x + 2)^2 + (y - 5)^2 = 9$:

- $(2, -5)$
- $(-2, 5)$
- $(2, 5)$
- $(-2, -5)$

Its radius:

- 9
- 3
- 81
- $\sqrt{3}$

A line meets a circle in at most:

- 1
- 2
- 3
- 4

$D = 0$ means:

- a secant
- a tangent
- no meeting
- a diameter

The chord at distance d:

- $\sqrt{r^2 - d^2}$
- $2\sqrt{r^2 - d^2}$
- $2d$
- $r + d$

AB equals:

- $OA - OB$
- $OB - OA$
- $OA + OB$
- $|AB|$

Collinear needs, beyond parallel:

- equal length
- a common point
- the origin
- nothing

An enlargement of factor k multiplies areas by:

- k
- k^2
- k^3
- $2k$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Coordinate method	Koordinatalar usuli	Координатный метод
Perpendicular bisector	O‘rta perpendikulyar	Серединный перпендикуляр
Discriminant	Diskriminant	Дискриминант
Tangent	Urinma	Касательная
Position vector	Radius-vektor	Радиус-вектор
Concept map	Tushunchalar xaritasi	Карта понятий
Target	Maqsad	Цель

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Q2 needs two things:

- two points
- the midpoint and the perpendicular gradient
- the length
- the radius

The faster route in Q5 is:

- the discriminant
- the distance from the centre
- guessing
- a sketch

A vector proof of collinearity ends with:

- the ratio
- a sentence naming the common point
- the magnitude
- nothing

Lessons 63–68 return to:

- circles
- solid geometry
- probability
- vectors in the plane

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $|AB|$ for $A(-1, 2), B(7, 8)$

   ANSWER 10

2. Midpoint of the same

   ANSWER $(3, 5)$

3. Gradient of the same

   ANSWER $\frac{3}{4}$

4. Centre of $x^2 + y^2 - 8x + 2y - 8 = 0$

   ANSWER $(4, -1)$

5. Radius of the same

   ANSWER 5

6. Gradient perpendicular to $\frac{3}{4}$

   ANSWER $-\frac{4}{3}$

7. PQ for $P(1, 2), Q(4, 8)$

   ANSWER $\begin{pmatrix} 3 \\ 6 \end{pmatrix}$

Medium · the standard question

7 PROBLEMS

1. Perpendicular bisector of $A(-1, 2)$ and $B(7, 8)$

ANSWER $4x + 3y = 27$

2. Chord of $x^2 + y^2 - 8x + 2y - 8 = 0$ on $y = x - 1$

ANSWER $2\sqrt{17} \approx 8.25$

3. Tangency values of c for $y = 2x + c$ and $x^2 + y^2 = 45$

ANSWER $c = \pm 15$

4. Are $P(1,2)$, $Q(4,8)$, $R(7,14)$ collinear?

ANSWER Yes — $QR = PQ$, Q common

5. Line through $(2, 3)$ parallel to $4x + 3y = 1$

ANSWER $4x + 3y = 17$

6. Distance from $(4, -1)$ to $y = x - 1$

ANSWER $\frac{4}{\sqrt{2}} = 2\sqrt{2}$

7. Circle centre $(4, -1)$ radius 5, in expanded form

ANSWER $x^2 + y^2 - 8x + 2y - 8 = 0$

Hard · stretch and reason

7 PROBLEMS

1. Tangent to $x^2 + y^2 = 45$ at $(6, 3)$

   ANSWER $6x + 3y = 45$, i.e. $2x + y = 15$

2. Circumcentre of $(-1, 2), (7, 8), (7, 2)$

   ANSWER $(3, 5)$

3. The line $4x + 3y = 27$ and the circle $x^2 + y^2 = 25$: how many points?

   ANSWER None — $d = 5.4 > 5$

4. Area of the triangle $(0, 0), (8, 0), (3, 6)$

   ANSWER 24

5. Find k so that $\begin{pmatrix} k \\ 9 \end{pmatrix}$ is parallel to $\begin{pmatrix} 2 \\ 6 \end{pmatrix}$

   ANSWER $k = 3$

6. Shortest distance from $(10, 5)$ to the circle $(x - 4)^2 + (y + 1)^2 = 25$

   ANSWER $\sqrt{72} - 5 \approx 3.49$

7. Show that the perpendicular bisectors of the sides of $(0,0), (6,0), (0,8)$ meet at $(3, 4)$

   ANSWER All three pass through it

07 Homework

Homework — 4 tasks

Bring the concept map to Lesson 63; the solid-geometry revision assumes the plane work is secure.

1. Rewrite in full every control-work question that lost a mark, naming the slip in the margin.
2. Complete the concept map with all six links written as sentences.
3. Find the values of c for which $y = x + c$ is a tangent to $x^2 + y^2 = 18$, by both routes.
4. Write your target for the revision block in one checkable sentence, and date it.